

# Swasthya Sahay Mitr

Healthcare Innovation | Your Pocket Public Health Companion

Category: AI-Powered Healthcare Platform | Languages: English & Hindi (Bilingual)

## Project Overview

Swasthya Sahay Mitr is a comprehensive, AI-powered healthcare ecosystem designed to bridge the critical gap between patients and medical services in India and beyond. Built with a focus on accessibility and cutting-edge technology, it serves as a digital health companion for users across diverse linguistic and geographic contexts, with native bilingual support in English and Hindi.

## Core Objectives

- Empowering Patients:** Providing instant, reliable medical guidance through conversational AI, enabling users to make informed health decisions.
- Universal Accessibility:** Ensuring healthcare information is available at every fingertip, regardless of location, connectivity, or language.
- Preventive Care:** Offering intelligent symptom-checking and health monitoring tools to detect and address issues before they escalate.
- Bilingual Reach:** Serving diverse populations with seamless English and Hindi (Bilingual) support throughout the platform.

## Key Features

|                               |                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SwasthAI Assistant            | A sophisticated conversational AI chatbot that answers medical queries, provides first-aid advice, and guides users on their personal health journey — available 24/7. |
| Intelligent Symptom Checker   | A diagnostic tool allowing users to input symptoms and receive preliminary health assessments (Emergency / Moderate / Mild) with clear, actionable recommendations.    |
| Nearby Healthcare Finder      | Map-integrated service using Leaflet.js and OpenStreetMap to locate the nearest hospitals, clinics, and pharmacies in real-time.                                       |
| Prescription & Report Scanner | Leverages OCR technology to scan and digitise physical medical prescriptions and lab reports for centralised, paperless management.                                    |

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|  <b>Women's Health Suite</b>             | A dedicated module featuring a Period Tracker, Pregnancy Care resources, and specialised health alerts (e.g., Anemia detection).                 |
|  <b>Health Dashboard &amp; Reminders</b> | A centralised hub for tracking medical history, appointment reminders, and medication schedules — keeping users on top of their health routines. |

## Technology Stack

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|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Frontend</b>         | React 19 with TanStack Start for robust routing and state management; Tailwind CSS v4 for a premium, glassmorphic UI.  |
| <b>Backend &amp; DB</b> | Supabase — secure user authentication, real-time database management, and serverless functions.                        |
| <b>Mapping</b>          | Leaflet.js integrated with the Overpass API for high-density, cost-effective healthcare facility mapping.              |
| <b>AI Integration</b>   | Seamless LLM integration powering the SwasthAI medical agent with contextual understanding and multilingual responses. |
| <b>OCR Engine</b>       | Optical Character Recognition pipeline for scanning and digitising prescriptions and medical reports.                  |
| <b>Deployment</b>       | Optimised for the edge using Cloudflare Pages — fast, globally distributed, and scalable.                              |

## Unique Value Proposition

*Unlike generic health apps, Swasthya Sahay Mitr combines AI-driven diagnostics with localised healthcare discovery. Its bilingual support (Hindi & English) and specialised focus on Women's Health make it a versatile tool for diverse populations. By merging intelligent symptom analysis, real-time facility mapping, and dedicated women's wellness modules, the platform ensures that Smarter Healthcare truly leads to Better Living.*

## Social Impact & Reach

- Rural Healthcare Access:** Reduces dependence on physical visits by delivering AI-powered medical guidance to anyone with a smartphone.
- Women's Empowerment:** Dedicated reproductive and maternal health tools address critical gaps in women's health literacy.

- **Language Inclusivity:** Hindi support ensures the platform reaches hundreds of millions of users in their native language.
- **Preventive Health Culture:** Early symptom assessment tools encourage proactive healthcare rather than reactive treatment.

## 🔗 Real-World Impact & Urgency

India faces one of the world's most acute healthcare access crises. With over 1.4 billion people and a doctor-to-patient ratio far below the WHO-recommended 1:1,000, millions of citizens — especially in rural and semi-urban areas — go without timely medical attention. Swasthya Sahay Mitr was built to directly address these systemic gaps.

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|--------------------|-----------------------------------------------------------------------------------|
| <b>600 M+</b>      | Indians lack reliable access to primary healthcare                                |
| <b>1 : 1,456</b>   | Doctor-to-patient ratio in India vs WHO's recommended 1:1,000                     |
| <b>70%</b>         | of disease burden in rural India is preventable with early intervention           |
| <b>56%</b>         | of women in India have never consulted a doctor for reproductive health           |
| <b>2nd Largest</b> | Hindi-speaking population globally — yet digital health tools remain English-only |

## 🔗 Why Now?

- **Post-Pandemic Digital Readiness:** COVID-19 accelerated smartphone penetration across India, with 750M+ active internet users in 2024. The infrastructure for digital health exists — what was missing was an accessible, trustworthy app in local languages.
- **Maternal & Women's Health Emergency:** India accounts for 11% of global maternal deaths. Lack of awareness, geographic distance, and cultural barriers prevent women from seeking timely care. The Women's Health Suite directly targets this crisis.
- **Overburdened Public Health System:** Government hospitals in India see 3–5x their intended patient load. AI-assisted triage and symptom checking can redirect non-critical cases, freeing up doctors for emergencies that truly need them.
- **Rising Non-Communicable Diseases (NCDs):** Diabetes, hypertension, and cardiac disease are surging in India — projected to cost \$4.58 trillion by 2030. Early symptom detection and health monitoring, core to this app, are proven cost-savers.
- **Last-Mile Connectivity:** With 5G and Jio's affordable data plans, even tier-3 towns now have connectivity. Swasthya Sahay Mitr is architected for low-bandwidth environments, ensuring it works where people need it most.

🔗 *Every minute of delayed diagnosis in a remote village can be life-changing. Swasthya Sahay Mitr is not just a health app — it is an equaliser. It gives a first-generation smartphone user in*

*rural Chhattisgarh the same quality of preliminary health guidance as someone walking into a private clinic in Mumbai. That is the urgency. That is the mission.*

***"Smarter Healthcare. Better Living."***